

Monday Meeting Brief

Cross-Patient Speech Decoding from Intra-Op μ ECoG

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Current Thesis

The bottleneck is **not** generic sequence modeling. It is **cross-patient spatial comparability** under variable array placement and partial cortical coverage. The current claim is:

- **Objective:** learn a patient-invariant latent of **cortical articulatory state**, not just a model that fine-tunes quickly.
- **v14 decomposition:** per-patient atlas-guided calibration, then shared spatiotemporal dynamics, then phoneme decoding.
- The shared object is an atlas-grounded regional token space, not raw channels.

What Changed Since Friday

The Friday PDF is now stale in five important ways:

- **No more spherical nodes / Wendland kernel as the default.** We now use **Brainnetome volumetric parcellation membership**.
- **No more single 16-node interface by default.** The active interface is **21 atlas/subparcel tokens**, with selective splitting of elongated motor/sensory parcels.
- **Stronger temporal front-end.** The old minimal Conv1d stem is replaced by a small **BaRISTA-style shared temporal tokenizer** over overlapping per-channel patches.
- **Spatial mapping is now atlas-grounded local summarization, not multi-view attention.** The default is a small **shared within-parcel Perceiver-style summarizer** that emits atlas/subparcel tokens; mean+gradient pooling is now the linear ablation.
- **Decoder is now a joint readout.** Three learned queries cross-attend to *all* token-time features with autoregressive conditioning, rather than relying on rigid temporal pooling.

Why This Direction Is Defensible

- **Best literature support:** BaRISTA is the strongest paper for our spatial question. Its central result is that **parcel-level encoding beats channel-level encoding** and also beats PopT-style channel-based baselines on key tasks.
- **Independent convergence:** MIBRAIN / Neuro-MoBRE also converge on **atlas or region-level common spaces** for cross-patient intracranial decoding.
- **Scale alone is not enough:** huge generic pretraining corpora still do *not* remove the need for explicit spatial identity. SWEC / MVPFormer are strategically important, but they do not beat more spatially grounded models on every Brain Treebank task.
- **Precision sharpens the architecture:** ultra-dense μ ECoG shows richer local structure exists inside parcels, so the right hierarchy is now explicit in the default model: **local within-parcel Perceiver summarizer** $\rightarrow$ **atlas/subparcel tokens** $\rightarrow$ **shared dynamics**. The remaining question is how small and shared that local summarizer should be.

Main Decisions Needed Monday

- **Data strategy:** Are we aligned that the highest-value request is **Flinker + Chang** for overt speech ECoG, while internal speech sEEG is the first SSL source and SWEC is only a staged generic pretraining source?
- **Anatomy / coordinates:** Can Zac confirm whether **T1 MRIs** exist for nonlinear MNI normalization? Current **talairach.xfm** affine residual is the main technical blocker for any spatial claim.
- **Model scope:** Are we aligned that the **within-parcel Perceiver summarizer is now the default v14 interface**, with mean+gradient as the main simplifying ablation rather than the other way around?

Success criteria: better held-out-patient transfer, graceful degradation under missing parcel coverage, and stronger alignment of homologous atlas tokens across patients. **Meeting goal:** leave with a locked first-pass plan for data acquisition, coordinate handling, and the smallest defensible default within-parcel Perceiver summarizer.